verilog代码整理

3-8译码器

```
module decoder3_8(y, a, g1, g2, g3);
  output[2:0] y;
  input[2:0] a;
  input g1, g2, g3;
  reg[2:0] y;
always @ (a, g1, g2, g3)
 begin
   if(g1 ==0) y = 8'b1111_1111;
    else if(g2 ==1) y = 8'b1111_1111;
    else if(g3 ==1) y = 8'b1111_1111;
   else
     begin
    y = 8'b0000_0001<<a;
           y = ~y;
     end
end
endmodule
```

超前进位加法器

```
module fulladd4(sum, c_out, a, b, cin);
  output [3:0] sum;
  output c_out;
  input [3:0] a, b;
  input cin;
  wire p0, g0, p1, g1, p2, g2, p3, g3;
  wire c4, c3, c2, c1;
  assign p0 = a[0] ^ b[0];
         p1 = a[1] ^ b[1];
         p2 = a[2] ^ b[2];
         p3 = a[3] ^ b[3];

  assign g0 = a[0] & b[0];
         g1 = a[0] & b[1];
         g2 = a[0] & b[2];
         g3 = a[0] & b[3];
```

```
  assign c1 = g0 | (p0 & cin),
         c2 = g1 | (p1 & c1) ,
         c3 = g2 | (p2 & c2),
         c4 = g3 | (p3 & c2);

  assign sum[0] = p0 ^ cin;
         sum[1] = p1 ^ c1;
         sum[2] = p2 ^ c2;
         sum[3] = p3 ^ c3;

  assign cout = c4;
endmodule
```

全减器

```
module sub4 (cin, dout, cout, a, b);
 output[3:0] dout;
 output cout;
  input[3:0] a, b;
  input cin;
  reg[3;0] dout,
  reg cout;
 always @  (a, b)
   begin
      {cout, dout} = a - b - cin;
   end
endmodule
```

```
module top(

  IN1   ,

  IN2   ,

  CLK   ,

  OUT   );

 input signed[7:0] IN1, IN2;

 input CLK;
```

```
 output signed [15:0] OUT;

 reg signed[15:0] OUT;

 reg signed[7:0] in1_d1R, in2_d1R;

 reg signed[15:0] mul_out;

always@(posedge CLK) begin // 生成D触发器的always块

  in1_d1R <= IN1;

  in2_d1R <= IN2;

  OUT     <= mul_out;

end

 always@(in1_d1R or in2_d1R) begin // 生成组合逻辑的always 块

  mul_out = in1_d1R * in2_d1R;

end

endmodule
```

计数器
清零，置数，计数 功能

```
module cnt16 (cout, q, clk, clr, load, en, d);
  output[3:0] q; //输出
  output cout; //进位信号
   input clk, clr, load, en;
   input[3:0] d;
   reg[3:0] q;
   reg cout;
  always @  (posedge clk)  begin
    if (clr) begin q <= 0; end
    else if (load) begin q <= d; end
            else if (en) begin
                 q <= q + 1;
                 if(q == 4'b1111) begin cout <= 1; end
                 else begin  cout <= 0; end
```

```
                    end
                    else begin q <= q; end
    end
endmodule
```

时钟

```
module count60(qout, cout, data, load, cin, reset, clk);
  output [7:0] qout;
  output cout;
  input [7:0] data;
  input load, cin, clk, reset;
  reg [7:0] qout;
always @ (posedge clk)
    begin
        if (reset) qout <= 0;
        else if (load) qout <= data;
        else if (cin) begin
            if(qout[3:0] == 9) begin
                qout[3:0] <= 0;
                   if(qout[7:4] == 5) qout[7:4] <= 0;
                   else  qout[7:4] <= qout[7:4]+1;
                     end
            else   qout[3:0] <= qout[3:0]+1;
        end
    end
  assign cout = ((qout == 8'h59)&cin)?1:0;
endmodule
```

异步计数器

```
module yb_cnt4(q, clk, rst);
   output[3:0] q;
   input clk, rst;
   reg[3:0] q;
   reg[3:0] qn;
always @ (posedge clk)
  begin if(!rst) begin q[0] = 0; qn[0] = 1; end
        else begin  q[0] = ~q[0]; qn[0] =~q[0]; end
  end
 always @ (posedge qn[0])
  begin if(!rst) begin q[1] = 0; qn[1] = 1; end
```

```
        else begin q[1] = ~q[1]; qn[1] =~q[1]; end
  end
 always @ (posedge qn[1])
  begin if(!rst) begin q[2] = 0; qn[2] = 1; end
        else begin q[2] = ~q[2]; qn[2] =~q[2]; end
  end
 always @ (posedge qn[2])
  begin if(!rst) begin q[3] = 0; qn[3] = 1; end
        else begin q[3] = ~q[3]; qn[3] =~q[3]; end
  end
endmodule
```

可变模计数器可以通过模值控制端来改变计数器的模值

```
module yb_cnt16(q, clk, clr, load, m);
   output[6:0] q;
   input clk, clr, load;
   input[6:0] m;
   reg[6:0] q;
   reg[6:0] md;
 always @ (posedge clk)
  begin    md <= m-1;
  begin if(!clr) begin q <= 0; end
        else begin
            if(load) begin q  <=  md; end
            else  begin
                if(q == md)
                  begin q <= 0; end
                else    begin q <= q + 1; end
end
        end
   end
  end
endmodule
```

寄存器

```
module reg8_1 (q, d, clk, oe);
  output[7:0] q;//数据输出
  input[7:0] d;  //数据输入
  input oe, clk;  //三态控制端，时钟信号
```

```
  reg[7:0] q;
 always @ (posedge clk)
   begin
      if(oe) begin
    q <= 8'bz;
      end
      else begin
            q <= d;
      end
  end
endmodule
```

锁存器

锁存器是一种与寄存器类似的器件。与寄存器采用同步时钟信号控制不同，锁存器是采用电位信号来进行控制的。

```
module reg8_1 (q, d, g, oe);
  output[7:0] q;//
  input[7:0] d;  //
  input oe, g;  //三态控制端，控制信号
  reg[7:0] q;
 always @ (*)
   begin
      if(oe) begin
    q <= 8'bz;
                end
      else begin
           if(g) q <= d;
             end
  end
endmodule
```

移位寄存器

```
module siso4 (dout, clk, din);//串入串出
  output dout;//
  input clk;
  input din;
  reg dout;
  reg[3:0] q;
always @ (posedge clk)
   begin
```

```
      q[0] <= din;
      q[3:1] <= q[2:0];
       dout <= q[3];
   end
endmodule
```

```
module sipo (dout, din,clr,clk);//串入并出
  output[4:0] dout;
  input clk, din,clr;
  reg[4:0] dout;  //五位
 always @ (posedge clk )
   begin
      if(clr) begin
    dout <= 0;
      end
      else begin
            dout <= {dout, din};
      end
endmodule
```

```
module piso4 ( dout, clk,clr,din);//并入串出
  output dout;//
  input clk, clr;
  input[3:0] din;
  reg dout;
  reg[1:0] cnt;  //初始为0
  reg[3:0] q;
always @ (posedge clk)
   begin
      if(clr) begin q <= 4'b0000; end
      else begin if(cnt > 0) begin q[3:1] <= q[2:0];
                     cnt <= cnt -1; end
                          else if(cnt == 2'b00)
                                    begin q <= din; cnt<=2'b11; end
                 end
       dout <= q[3];
   end
endmodule
```

```
module d_reg (dout_l, dout_r, clk, din, left_right);
  output dout_l, dout_r;
```

```
  input clk, din, left_right;
  reg dout_l, dout_r;
  reg[7:0] q_tempt; //内部有八位
  integer i;

always @ (posedge clk) //双向移位寄存器
   begin
      if(left_right) begin
           q_tempt[7] <= din;
            for (i = 7; i >= 1; i = i - 1)
          begin q_tempt[i-1] <= q_tempt[i]; end
      end
      else begin
    q_tempt[0] <= din;
            for (i = 1; i <= 7; i = i + 1)
          begin q_tempt[i] <= q_tempt[i-1]; end
       end
     dout_r <= q_tempt[0];
     dout_l <= q_tempt[7];
    end
endmodule
```

同步复位的D触发器

```
module flipflop (D, Clock, Resetn, Q);
    input D, Clock, Resetn;
    output Q;
    reg Q;
    always @(posedge Clock)
        if (!Resetn)
            Q <= 0;
        else
            Q <= D;
endmodule
```

异步复位的D触发器

```
module flipflop (D, Clock, Resetn, Q);
   input D, Clock, Resetn;
   output Q;
   reg Q;
   always @(negedge Resetn or posedge Clock)
      if (!Resetn)
         Q <= 0;
      else
         Q <= D;
endmodule
```

分频器

```
module div248(div2,div4,div8,clk);
  output div2, div4, div8;
  input clk;
  reg div2, div4, div8;
  reg[2:0] cnt; //初始化
always @ (posegdge clk)
    begin
      cnt <= cnt+1;
      div2 <= cnt[0];
      div4 <= cnt[1];
      div8 <= cnt[2];
    end
endmodule
```

```
module clkdiv (
input  mclk,   //50MHz
input  clr,
output clk190,
```

```
output clk48
);
reg [24:0] q;  // 25 位计数器
always @ (posedge mclk or posedge clr)
    begin
       if (clr == 1)
               q <= 0;
       else
              q <= q + 1;
    end
   assign clk190 = q[17]; // 190 Hz
   assign clk48 = q[19]; // 47.7 Hz
endmodule
```

周期不是2的次方

```
module div6(div6, clk);
  output div6;
  input clk;
  reg div6;
  reg[2:0] cnt;
 always @ (posedge clk)
    begin
      if (cnt == 3'b010) begin//半个周期时
            div6 <= ~ div6;
             cnt <= 0;
          end
      else begin
              cnt <= cnt + 1;
              end
    end
endmodule
```

占比不是1:1

```
module div6(div6, clk);
  output div6;
  input clk;
  reg div6;
  reg[2:0] cnt;
 always @ (posedge clk)
    begin
```

```
      if (cnt == 3'b101) cnt <= 0;
     else cnt <= cnt + 1;
    end
 always @ (posedge clk)
    begin
      if (cnt == 3'b000) div6 <= 0;
      if (cnt == 2'b010) div6<=1;
    end
endmodule
```

```
module  vote7 ( pass,vote );
      output pass;
      input [6:0] vote;
      reg[2:0] sum;      //sum为reg型变量，用于统计赞成的人数
      integer i;
      reg pass;
      always @(vote)
          begin
              sum = 0;                                    //sum初值为0
              for(i = 0;i<=6;i = i+1)                 //for语句
        if(vote[i])      sum = sum+1;
                                                        //只要有人投赞成票，则
sum加1
              if(sum[2])     pass = 1;  //若超过4人赞成，则表决通过    else
pass = 0;
          end
endmodule
```